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Complete remagnetization of some Permian dykes from western Norway induced during burial/uplift

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Stable paleomagnetic directions in four basaltic dykes and in some associated Caledonian metamorphic rocks define high-latitude, Mesozoic paleomagnetic pole positions which are not compatible with the K–Ar age of ~250 My determined for the dykes. A monotonic increase of ~40% occurs in the potassium content of samples taken across a dyke 32 cm wide. This is accompanied by only a 1–2% variation in the K–Ar age of the samples, suggesting the absence of any significant level of initial argon. Titanomagnetite grains having bulk compositions around $x = 0.6$ have suffered extensive low-temperature alteration, forming assemblages of ferri-rutile granules in a matrix of pure magnetite. The complete remagnetisation of both the dykes and the associated country rocks is probably an expression of a VRM acquired at elevated temperatures (150–500°C) at the emplacement depth of the dykes. A stable remanent magnetisation was locked-in during uplift of the area, probably related to the Kimmerian basin development in the adjacent North Sea and the epeirogenic uplift of western Fennoscandia.

1. Introduction

Magnetic overprinting may be recognised by the identification of paleomagnetic components which can be demonstrated to be significantly younger than the geological unit concerned. In the case of partial overprinting, demagnetisation experiments may uncover two significantly different paleomagnetic components within even a single specimen. In such a case it can be inferred, without recourse to isotopic age studies of the rock, that one of the components represents a younger geomagnetic field direction. By contrast, however, complete magnetic overprinting, reflected by a single, stable magnetic component, can be demonstrated only if both the age of the geological unit and that of its remanent magnetisation can be determined unambiguously. The isotopic evolution and paleomagnetic properties of a rock both depend critically on its thermal history. Thus, it is likely that any thermal process which resets the initial remanent magnetisation may also influence any isotopic age determination. In the present case the isotopic composition across a thin diabase dyke strongly suggests the absence of any initial argon, as well as freedom from major argon loss. The Permian K–Ar age obtained, however, is incompatible with the Mesozoic pole position defined by the single-component, stable magnetisation which is observed. The possible origin of this complete magnetic overprint is discussed.

2. Geology and sampling

On the island of Sotra, western Norway, N–S-striking diabase dykes (intruding Caledonian metamorphic rocks) were sampled at nine locali-

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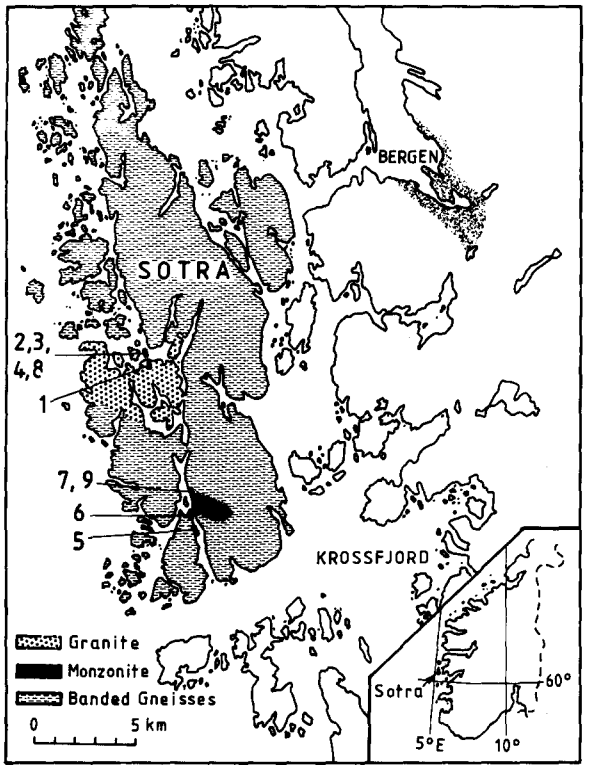


Fig. 1. Geological sketch map of the island of Sotra showing sampling localities.

ties (see Fig. 1). The exposed width of the fresh-looking dykes varies between 0.3 and 0.75 m. The geochemical composition (site 2) is characteristic of continental, tholeiitic basalts (Furnes, personal communication, 1981). There are no well-developed glassy or chilled margins, nor has any post-intrusive tectonism been recognised in the area.

In addition to the oriented (sun/magnetic compass) drill cores collected from the diabase dykes, associated country rocks (assumed to have been baked) were sampled at five sites at distances up to 2.5 times the dyke widths.

3. Radiometric age determination

Whole-rock K–Ar analyses were performed for seven diabase samples collected at approximately equal intervals across a 32-cm wide section of the dyke at site 8, and for two additional samples from site 5. The K–Ar age data are summarised in Table I and show that, with the exception of one sample from site 8, which yielded an age of 228 My, all determinations were confined to a narrow age range between 249 and 271 My, six being concordant at the 2σ level, with a mean age of 262 ± 6 My.

A steady increase in both the potassium and argon contents is observed from east to west across

TABLE I

K–Ar age data summary

Collection reference	Sample location [a]	K_2O	Radiogenic Ar (10^{-2} mm^3 g^{-1})	Atmospheric Ar (%)	Age (My)
	Site 8				
SOK-1	W contact	2.07±0.03	1.62 ±0.02	5.0	228±5
SOK-2	E contact	2.18±0.03	1.97 ±0.02	3.2	260±4
SOK-3	10 cm W	2.45±0.02	2.11 ±0.02	2.4	249±3
SOK-4	4 cm W	2.34±0.03	2.09 ±0.02	1.9	258±4
SOK-5	21 cm E	2.50±0.01	2.27 ±0.03	2.7	262±4
SOK-6	12 cm E	2.52±0.01	2.38 ±0.03	3.2	271±4
SOK-7	3 cm E	2.92±0.02	2.73 ±0.03	2.8	268±3
	Site 5				
SO18-A	15 cm W	1.27±0.02	1.099±0.005	18.8	250±3
SO19-A2	5 cm W		1.136±0.005	16.1	257±4

[a] Site 8, dyke 32 cm wide; site 5, dyke 50 cm wide. $\lambda_e = 0.581 \times 10^{-10}$ y^{-1}; $\lambda_\beta = 4.962 \times 10^{-10}$ y^{-1}; $^{40}K/K = 1.167 \times 10^{-2}$ atom %.

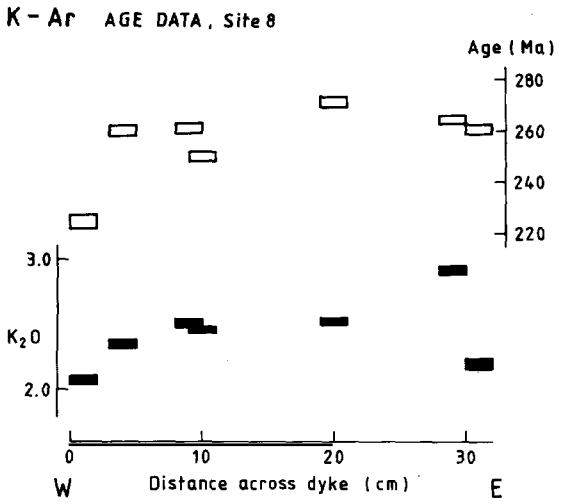


Fig. 2. Variation of potassium content and age determinations for seven samples across a dyke 32 cm wide at site 8.

site 8 (see Fig. 2 and Table I). Although the potassium content varies by some 40%, the associated ages change by only a few percent. The oldest ages are found in samples having the highest potassium contents, and this feature is inconsistent with any model of assimilation of a uniform level of initial argon as being a factor possibly influencing the age determinations. It is regarded as highly unlikely that any level of assimilated initial argon should be precisely sufficient to maintain such a constant age even though the radiogenic contribution varies by some 40%. Equally, we would argue that argon loss is unlikely to have influenced the K–Ar age since concordance exists among six of the eight samples analysed. Any "isotopic" overprint which may have occurred would have had to have been a total one, and the "age" of closure of the dykes with respect to argon diffusion occurred at 262 ± 6 My, although the possibility that the dykes are even older cannot be excluded.

4. Paleomagnetic directions

All diabase samples were subjected to progressive thermal or alternating-field (AF) demagnetisation. Both treatments revealed the presence of a single, stable, downpointing magnetisation component, associated with smoothly decreasing intensity curves after AF demagnetisation (MDF > 20 mT) and almost square-shaped demagnetisation curves after thermal treatment, indicating a rather narrow range of blocking temperatures between 410 and 550°C (see Fig. 3). Some of these demagnetisation curves exhibit conspicuous "shoulders" at ~450°C, indicative of bimodal distributions of blocking temperature (see SO18, Fig. 3), which are not, however, associated with directional deviations.

The almost circular distribution of stable directions (Fig. 4) does not differ significantly from the distribution of NRM directions, indicating the absence both of VRM acquisition in the present field and of induced stray components associated with sampling and handling. The site directions define a mean of $D = 19.5°$, $I = 59.1°$ ($\kappa = 487$, $\alpha_{95} = 2.3°$) (Fisher, 1953), representing a paleomagnetic pole position situated at 66°N 145°E, which is close to the Jurassic–Cretaceous section on the

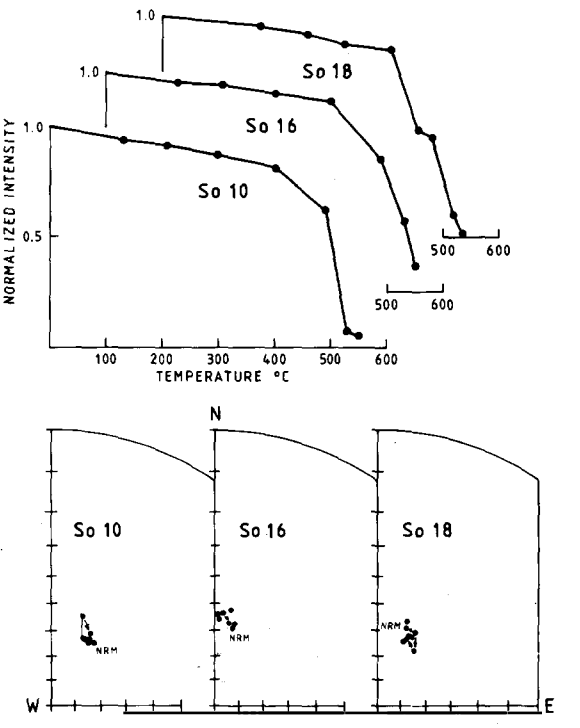


Fig. 3. Thermal demagnetisation curves and stereographic projections for three dyke specimens.

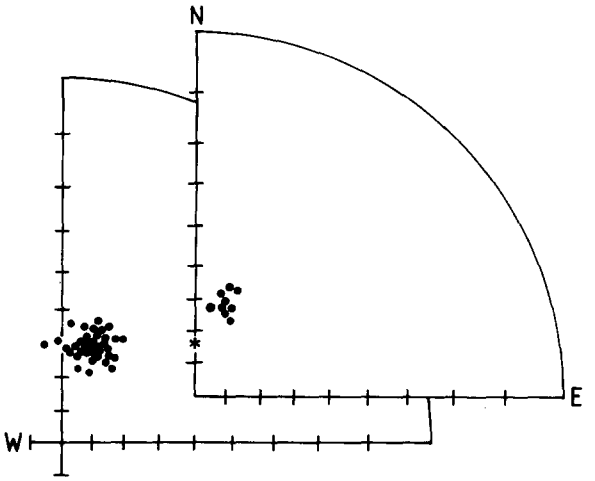


Fig. 4. Distribution of stable specimen directions (*left*) and site mean directions (*right*).

apparent wander path for Eurasia (Irving, 1977).

This inferred Mesozoic paleomagnetic pole is evidently inconsistent with the Permian K–Ar age obtained from the dyke material (during Permian times the paleomagnetic pole was situated at 45°N 163°E (Irving, 1977)). In addition to this major inconsistency between the paleomagnetic pole and the K–Ar age data, the single-component and normal magnetisation residing in the dykes is incompatible with the recognised worldwide reversed polarity (Kiaman epoch) of the Permian paleomagnetic field (McElhinny, 1973).

5. Rock magnetic properties

5.1. Ore microscopy

Microscopy revealed cubic Fe–Ti oxide grains, with skeletal to almost euhedral crystal shapes, having grain sizes predominantly in the range 10–80 μm. The titanomagnetite grains do not contain any high-temperature exsolved ilmenite lamellae. However, all grains have been exposed to low-temperature alterations, ranging from varying degrees of maghemitisation to the formation of ferri-rutile granulation (Ade-Hall et al., 1971), which may cover the grains completely. Tiny sulphide (?) grains, too small for proper identification, are also present.

5.2. Electron-microprobe analysis

The compositions of titanomagnetite grains in four diabase samples are presented in Table II. Attempts to discriminate between the composition of individual ferri-rutile granules and that of the surrounding matrix were not successful, probably owing to the small dimensions of the granules compared to the larger effective diameter of the electron beam. Hence, the observed cation compositions represent in effect bulk compositions of the homogeneous, primary crystallised titanomagnetite grains. Estimates of the approximate stoichiometric compositional parameters $x(Fe_{3-x}Ti_xO_4)$ range between 0.59 and 0.68 using the observed Fe/Ti ratios only. These Fe/Ti ratios will be on the high side taking into account that the original titanomagnetites also contain cations such as Mg, Al, Mn. This compositional range is characteristic of primary titanomagnetites crystallising from a basaltic (tholeiitic) magma (Hargraves and Petersen, 1971).

5.3. Thermomagnetic curves

The thermomagnetic curves for the diabase samples have special peculiarities, in that they all show, irrespective of sample position within the

TABLE II

Microprobe analysis of titanomagnetite grains

Sample	SO2	SO19	SO26	SO37
Site	1	4	5	8
N	19	14	12	9
MgO	1.61	0.60	0.93	0.31
Al_2O_3	2.42	1.94	0.79	2.92
Cr_2O_3	0.03	0.02	0.04	0.02
MnO	1.69	0.26	0.33	0.92
TiO_2	23.18	24.11	22.38	21.60
FeO	60.68	63.46	70.14	67.98
x [a]	0.68	0.68	0.59	0.59

[a] Compositional parameter for stoichiometric titanomagnetite ($Fe_{3-x}Ti_xO_4$), using Fe/Ti ratios only.

dykes, a single, initial and reversible Curie point of 580°C, indicative of pure magnetite. Upon cooling from 700°C in air, a few samples develop an additional and dominant reversible magnetic phase, having a Curie point of ~440°C. The composition of this phase has not been determined, but it is tentatively suggested that it reflects a partly rehomogenised titanomagnetite phase ($x \simeq 0.2$) caused by diffusion of titanium from the enriched granules into the Ti-deficient matrix.

A homogeneous titanomagnetite phase having a compositional parameter equivalent to that observed for these samples ($x = 0.59$–0.68) would have a Curie temperature less than 150°C. The observed pure-magnetite ($x = 0$) Curie point of 580°C indicates that an almost complete diffusion of all titanium out of the initial titanomagnetite structure has occurred, forming granules of ferri-rutile within a matrix of pure magnetite. The suggested magnetite matrix is favoured in contrast to the more straightforward interpretation involving the creation of a titanomaghemite phase because the saturation magnetisation after heating in air to 700°C decreases by only 5–10%. This observation is not compatible with thermomagnetic results obtained for synthetic titanomaghemite samples (Readman and O'Reilly, 1970), which exhibit a large (> 50%) decrease in saturation magnetisation after heating.

Low-temperature formation of the ferri-rutile granules probably took place at temperatures below 500°C, above which temperature deuteric oxidation is likely to produce exsolution of ilmenite lamellae (Petersen, 1976).

5.4. Cooling NRM

The intensity variation of the remanent magnetisation upon cooling to −196°C in a low ambient field (< 50 nT) revealed an abrupt and substantial (> 50%) decrease around −150°C. This coincides closely with the temperature at which the magnetocrystalline anisotropy constant for magnetite, K_1, passes through zero, indicating that a major proportion of the NRM is carried by almost pure magnetite.

The rock magnetic properties investigated indicate that the primary titanomagnetites ($x \simeq 0.63$) have been exposed to low-temperature alterations forming an assemblage of pure magnetite and ferri-rutile granules. The Lowrie–Fuller test (Lowrie and Fuller, 1971) demonstrates that the magnetite grains carrying the remanence are at least in part in a single-domain state.

6. Baked country rocks

A total of 30 country rock samples (gabbro and baked gneiss) collected at various distances from the dykes were subjected to progressive thermal-demagnetisation treatment. NRM directions remained stable to ~550°C, above which temperature laboratory-induced viscous magnetisation appeared, preventing further demagnetisation. The distribution of stable directions, shown in Fig. 5, is slightly elongated along the E–W direction, and defines a mean of $D = 1.0°$, $I = 53.4°$ ($\kappa = 17$, $\alpha_{95} = 6.7°$) (Fisher, 1953). The mean is very close to, but nevertheless significantly different from (at the 95% confidence level) the mean for the dykes which are assumed to have baked these rocks.

The absence of any directional trends during demagnetisation suggests either that only one magnetic component resides in the country rocks, or alternatively that the experimental techniques are not able to resolve the expectedly composite NRM direction into the pre-baking component and the assumed overprint induced during baking. For

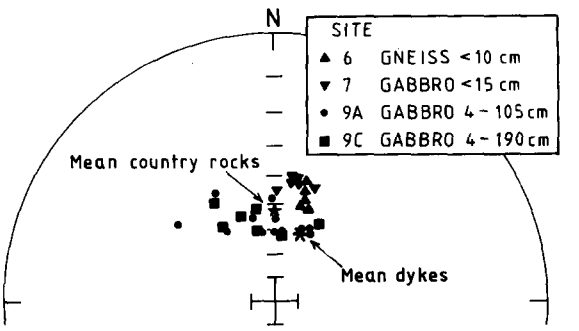


Fig. 5. Distribution of stable directions encountered in country rocks sampled within 2.5 times dyke width at three localities.

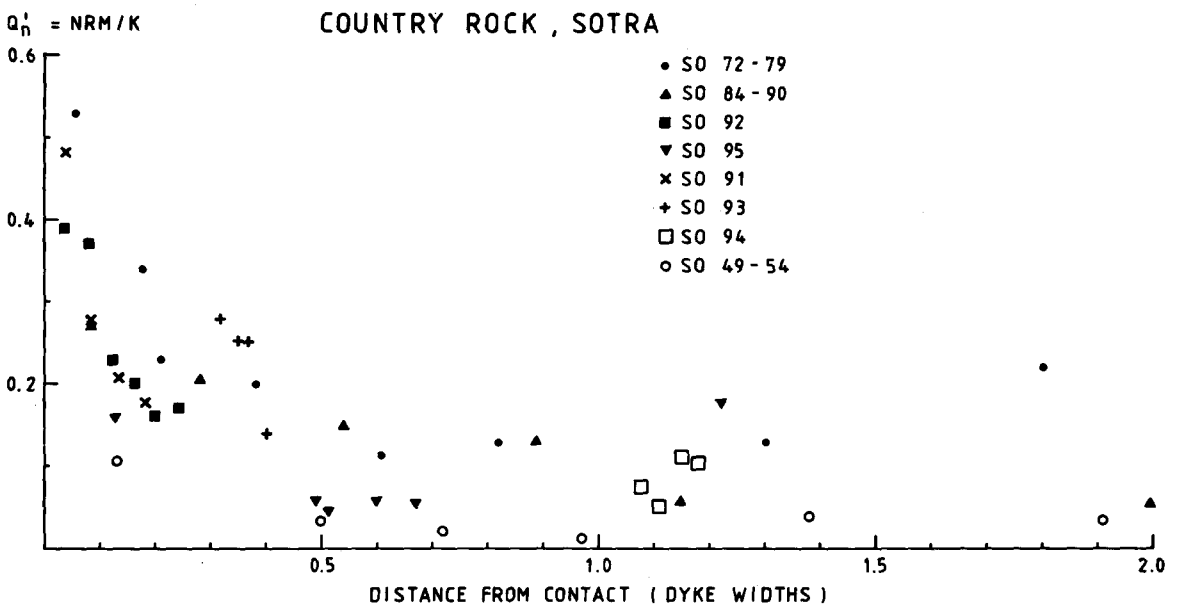


Fig. 6. Variations of modified Königsberger ratio ($Q'_n = \mathrm{NRM}/\kappa$) with distance from dyke contacts.

samples collected at some distance from the dykes, the onset of viscous behaviour in the laboratory occurs at temperatures well above the theoretical reheating temperatures associated with the intrusion of basaltic dykes. Theoretical considerations (Jaeger, 1968) suggest that at 2.5 times dyke width the country rock would be exposed to reheating temperatures of $\sim 1/5$ the temperature of the intruding magma. In the case of a basaltic magma ($T \simeq 1100°C$) this implies that any induced PTRM component caused by reheating at such a distance ought to be removed by demagnetisations above 230°C. The origin of the stable, single-component magnetisation observed in the country rock is thus concluded not to be associated with the intrusion of the dykes.

A possible effect of the dykes is, however, the establishment of a recognisable pattern in the variation of the modified Königsberger ratio ($Q'_n = \mathrm{NRM}/\kappa$) with distance from the dykes (see Fig. 6). A pronounced decrease of Q'_n by almost $1/3$ within a distance of one-half a dyke width may well reflect thermally induced alterations of the magnetic minerals.

7. Discussion and conclusion

The stable, single-component remanent magnetisation residing in both the K–Ar-dated Permian basaltic dykes and in sections of the adjacent country rock are not compatible with a Permian Kiaman reversed geomagnetic field direction. Since there is evidence against the existence of "initial" argon contamination in the dated samples, the age obtained represents, at worst, a minimum age. We are led to conclude, therefore, that both the dykes and country rocks must have acquired their present remanence in a Middle–Late Mesozoic paleomagnetic field.

The rock magnetic and mineralogical properties indicate that the primary crystallised titanomagnetite grains ($x \simeq 0.63$) in the dykes have been exposed to extensive low-temperature alterations, causing the formation of a major assemblage of ferri-rutile granules and pure magnetite. The absence of exsolved ilmenite lamellae indicates that this process took place at temperatures below 500°C. Secondary-mineral stability-field calibration of subaerial Tertiary lavas (Ade-Hall et al.,

1971) has indicated that the formation of granulation features occurs at temperatures above 150°C. Post-deuteric reheating to such elevated temperatures is unlikely, since the dykes investigated represent the youngest known magmatic activity in the area. Hence, it is suggested that the dykes were emplaced at ambient temperatures between 150 and 500°C, corresponding to depths of 5–15 km using a geothermal gradient of 30°C km^{-1}.

The initial Curie temperatures of the homogeneous primary titanomagnetite grains, as deduced from the observed bulk compositions of the granulated grains, were thus lower than the inferred ambient emplacement temperature. The acquisition of a stable chemical remanent magnetisation may thus have occurred in conjunction with an increase in Curie temperature to ~ 580°C associated with the formation of the ferro-rutile/magnetite assemblage. However, it is regarded as highly unlikely that both the basaltic dykes, banded gneisses and the rather acid gabbro (monzonite) should, by chemical alterations of the quite different magnetic mineralogies, acquire almost coinciding directions. It is therefore suggested that the origin of the complete magnetic overprint both of the dykes and of the associated country rocks is not related to chemical changes of the rocks. It is instead tentatively concluded that the secondary magnetisation represents a viscous remanent magnetisation acquired at ambient temperatures between 150 and 500°C, at depths of ~ 5–15 km, the temperature being sufficiently low not to induce appreciable argon loss. It is emphasised that the VRM interpretation is mineralogically unrecognisable, but that the alteration state and properties of magnetic minerals convey crucial information regarding the thermal history of the rocks. We suggest that the locking-in of the stable directions is related to a substantial decrease in temperature associated with uplift of the general area. This probably took place in conjunction with the Kimmerian development of the adjacent North Sea basin in Jurassic times, although a major epeirogenic uplift of western Fennoscandia is reported to have occurred in early Tertiary times (Egeberg and Eldholm, 1977).

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